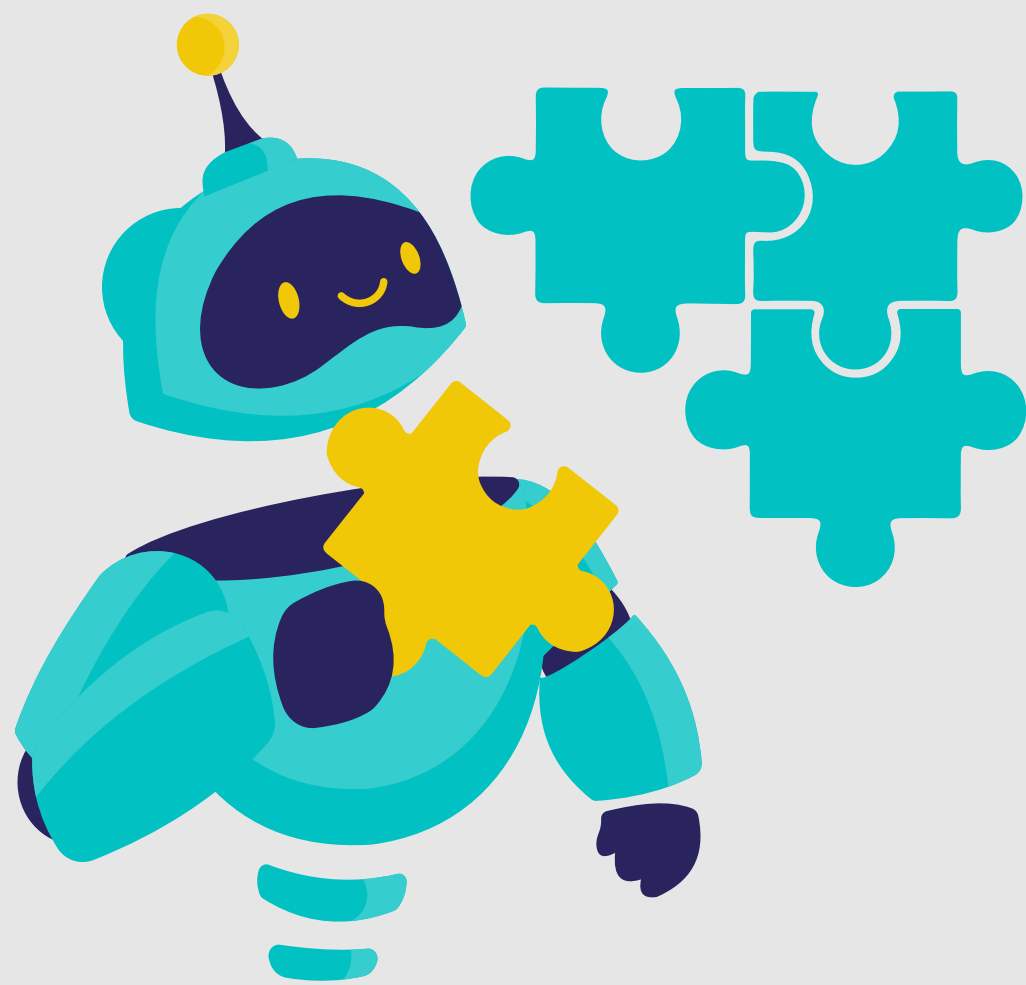


ASPECT—GLOBAL EVALUATIVE DISSONANCE IN ONLINE PRODUCT REVIEWS: A TEMPORAL ANALYSIS OF CONSUMER EXPERIENCE AND PRIOR REVIEW ENVIRONMENTS

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OUTLINE

- IDENTIFY ASPECT–GLOBAL EVALUATIVE DISSONANCE
- EXAMINE HOW CONSUMERS’ EVALUATIONS OF SPECIFIC PRODUCT ASPECTS CAN DIFFER FROM THEIR OVERALL PRODUCT EVALUATION.
- EXAMINE THE ROLE OF PRIOR REVIEW ENVIRONMENTS
- ANALYSE WHETHER THE OPINIONS, RATINGS, AND SENTIMENT IN EARLIER REVIEWS INFLUENCE HOW CONSUMERS EVALUATE PRODUCT ASPECTS IN SUBSEQUENT REVIEWS.
- ANALYSE THE TEMPORAL RELATIONSHIP
- INVESTIGATE HOW THE RELATIONSHIP BETWEEN ASPECT-LEVEL EXPERIENCES AND GLOBAL EVALUATIONS CHANGES OVER TIME AS THE REVIEW ENVIRONMENT EVOLVES.
- DEVELOP AND VALIDATE THE AGED FRAMEWORK
- QUANTIFY PATTERNS OF ASPECT–GLOBAL DISSONANCE AND DETERMINE WHETHER PRIOR REVIEW ENVIRONMENTS SYSTEMATICALLY EXPLAIN VARIATIONS IN THIS DISSONANCE, CONTRIBUTING TO A MORE DYNAMIC UNDERSTANDING OF ONLINE CONSUMER EVALUATION.

Litrature review

S. No.	Author(s), Year	Method / Approach	Key Contribution	Gap Identified
1	Lee, Tan, and Hosanagar (2015)	Multilevel ordered logistic regression, generalized linear latent variable and	Demonstrated that herding behavior occurs as higher predecessor ratings	Did not isolate individual-level heterogeneity, homophily, or the role of
2	Heinrich, Schiller, and Weinfurtner (2025)	Linear regression models and Mann-Whitney U tests on eight extensive real-	Proved that initial average ratings strongly influence long-term average ratings (77–	The long-term influence of initial star ratings on subsequent rating evolution had
3	Wang, Wang, Zhou, and Liu (2026)	Three behavioral experiments (simulated recall, dining evaluation platform) and	Showed that a "Rating-First" sequence polarizes overall consumer ratings	Unexplored how specific interface review sequences ("Rating-First" vs. "Review-
4	Cheng, Gao, Zhang, and Li (2025)	Cross-platform difference-in-differences (DID) approach comparing hotel reviews	Found that introducing GAIRS reduces subsequent review volume and length	Prior research focused on GAI as an external knowledge source or creator tool,
5	Wang, Chau, Peng, et al. (2022)	OLS regression using longitudinal secondary data of 127,133 reviews from US	Introduced anchoring effect theory to show that prior average ratings serve as	Prior research generally failed to examine how culture matters to the social influence
6	Matakos and Tsaparas (2016)	Empirical analysis of temporal metrics (mean, variance, deviation, kurtosis) and a	Proved that reviews tend to become more polarized (U-shape/J-shape) over time;	Existing herding models fail to explain observed bimodal distributions, and no
7	Li, Zhang, Meng, and Zhang (2019)	Ordered Logit and OLS regression of paired online restaurant reservations and	Found that review temporal distance has a direct negative effect on restaurant	Prior studies assumed immediate review posting and ignored how review temporal
8	Schlosser (2005)	Lab experiments/surveys exploring the impact of prior reviews on posting	Found that posters tend to adjust their evaluations downward after reading	Focused on how negative bias theory and differentiation effects operate in posting
9	Li and Hitt (2008)	Empirical panel analysis of Amazon book review data.	Demonstrated that early buyers have positive, idiosyncratic preferences (self-	Did not model sequential observational learning or separate social herding from
10	Wu and Huberman (2008)	Empirical analysis of online rating timelines using an opinion formation under	Found that average ratings converge to the mean in the short term; over time, only	Did not consider the long-term persistence of initial rating levels or multi-platform
11	Moe and Trusov (2011)	Empirical modeling of social dynamics and sales in a beauty product rating forum.	Showed that online product rating social dynamics (selection and adjustment	Did not explore the role of aggregate cultural factors or psychological distance
12	Godes and Silva (2012)	Econometric modeling of the sequential and temporal dynamics of book reviews.	Found that average ratings decline with rating ordinality rather than with time, due	Ignored how the actual temporal delay (in days/minutes) between product
13	Ho et al. (2017)	Expectation-disconfirmation structural modeling of online reviews.	Demonstrated that users are highly likely to review when their experienced quality	Focused primarily on pre-purchase expectation formulation without
14	Guo and Zhou (2016)	Empirical regression analysis of restaurant reviews.	Prior rating volume and variance moderate and mitigate the positive influence of prior	Relied strictly on social influence theory, leading to a paradox where high review

Litrature review

15	Sridhar and Srinivasan (2012)	Empirical regression of hotel ratings and features under a social influence	Found that existing average ratings moderate how specific features of hotel	Ignored how cross-cultural factors shape reviewers' susceptibility to this
16	Ma et al. (2013)	Empirical modeling of prior reviews in a reputation system.	Identified that consumers build expectations on prior average ratings,	Did not analyze the temporal aspects (such as review temporal distance) or the
17	Lederrey and West (2018)	Quantitative analysis of herding effects using natural experiments on product	Measured short-term herding effects, finding that a "high", "medium", or "low"	Focused only on the short-term impact of a single very first rating, neglecting the
18	Park, Shin, and Xie (2021)	Empirical analysis of a small data sample (less than 300 items) over a three-year	Observed that the very first rating (positive/negative) significantly impacts	Did not provide a theoretical framework (like bandwagon/differentiation) and
19	Filippas, Horton, and Golden (2022)	Empirical analysis of rating inflation across multiple online marketplaces.	Documented that user reviews and ratings tend to become systematically	Did not tie this inflation back to cognitive heuristics or platform evaluation
20	He and Bond (2015)	Behavioral/empirical analysis of dispersion in online word of mouth.	Found that high controversy (dispersion) of online reviews increases consumer	Did not explore how initial controversy persists and impacts the long-term rating
21	Duan, Gu, and Whinston (2008)	Empirical econometric analysis of panel data on online software reviews.	Found that while user reviews affect sales, users are also strongly guided by	Did not explore the long-term influence of the initial star ratings or differentiate
22	Xie, Chen, and Wu (2016)	Empirical analysis of hotel reviews and offline hotel popularity.	Proved that online users rely heavily on the most recent reviews as they contain	Did not analyze how initial rating features (such as trend and controversy) continue
23	Mudambi and Schuff (2010)	Empirical study of customer reviews on Amazon using regression models.	Determined that review extremity, text length, and depth determine review	Treated ratings and reviews strictly as input cues for consumers, ignoring how
24	Schlosser (2011)	Experimental study of review argumentative structures.	Demonstrated that including both pros and cons in the review argumentative	Focused on review consumption rather than how the sequence of writing reviews
25	Lee and Lin (2022)	Experimental study on online review presentation order.	Examined how review sorting methods (such as helpfulness vs. chronological)	Focused exclusively on the consumer side, leaving a gap on how generator-side
26	Hennig-Thurau et al. (2004)	Empirical survey of users on consumer-opinion platforms.	Identified the psychological and social motivations of consumers for writing	Did not address how platform interaction interfaces (such as evaluation sequences)
27	Auh, Salisbury, and Johnson (2003)	Quantitative customer satisfaction survey modeling.	Demonstrated order effects in customer satisfaction surveys: having an overall	Did not study the sequence between textual, unstructured reviews and
28	Eryarsoy and Piramuthu (2014)	Experimental evaluation of sequential bias in review presentation.	Found that the presentation order of reviews triggers sequential bias, leading	Unexplored sequential effects between a user's own textual expression and
29	Mehr and Simmons (2024)	Behavioral experiments on global and feature-specific evaluations.	Showed that rating specific features of an experience before providing a global	Focused strictly on numerical evaluations of sub-features without investigating the
30	Duffek, Merlo, Hoyer, and Eisingerich (2024)	Behavioral experiments comparing "Review-First" vs. "Rating-First" scenarios.	Provided direct evidence that writing a review before rating leads to different	Failed to control for the quality of service context (high vs. low quality), which alters

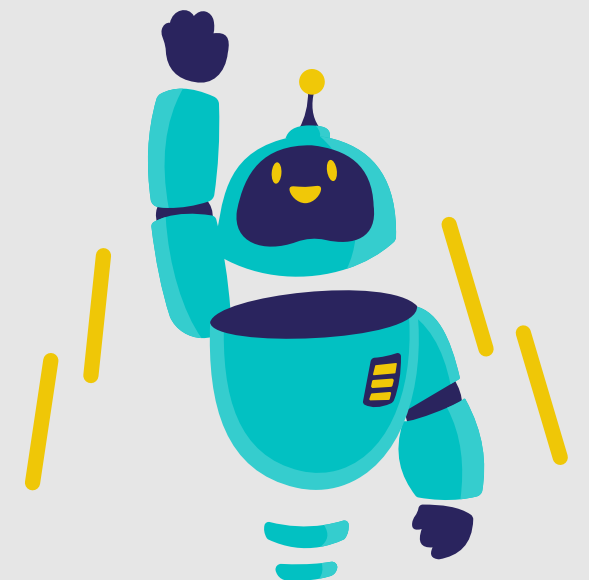
SCOPE & PROBLEM STATEMENT

- EXISTING RESEARCH STUDIES ASPECT-LEVEL EVALUATIONS, GLOBAL EVALUATIONS, AND PRIOR REVIEW INFLUENCE LARGELY IN ISOLATION.
- IT REMAINS UNCLEAR HOW THE PRIOR REVIEW ENVIRONMENT SHAPES THE RELATIONSHIP BETWEEN A CONSUMER'S CURRENT ASPECT-LEVEL EXPERIENCE AND OVERALL PRODUCT EVALUATION.
- THE ROLE OF ASPECT-LEVEL MISMATCH/DISSONANCE BETWEEN PRIOR REVIEWS AND CURRENT EXPERIENCE IN SHAPING GLOBAL EVALUATION REMAINS UNDEREXPLORED.



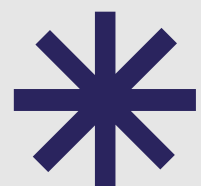
RESEARCH CHALLENGES

- ACCURATELY IDENTIFYING WHEN ASPECT-LEVEL OPINIONS CONFLICT WITH THE OVERALL PRODUCT EVALUATION.
- IDENTIFYING PRODUCT ASPECTS AND THEIR ASSOCIATED SENTIMENT FROM UNSTRUCTURED, DIVERSE REVIEW TEXT.
- DEFINING AND MEASURING WHAT A CONSUMER WAS EXPOSED TO IN EARLIER REVIEWS WITHOUT CONFUSING PRIOR INFLUENCE WITH GENERAL PRODUCT TRENDS.
- SEPARATING GENUINE CHANGES IN CONSUMER EVALUATION OVER TIME FROM CHANGES CAUSED BY PRODUCT UPDATES, REVIEW COMPOSITION, OR OTHER EXTERNAL FACTORS.



RESEARCH OBJECTIVES

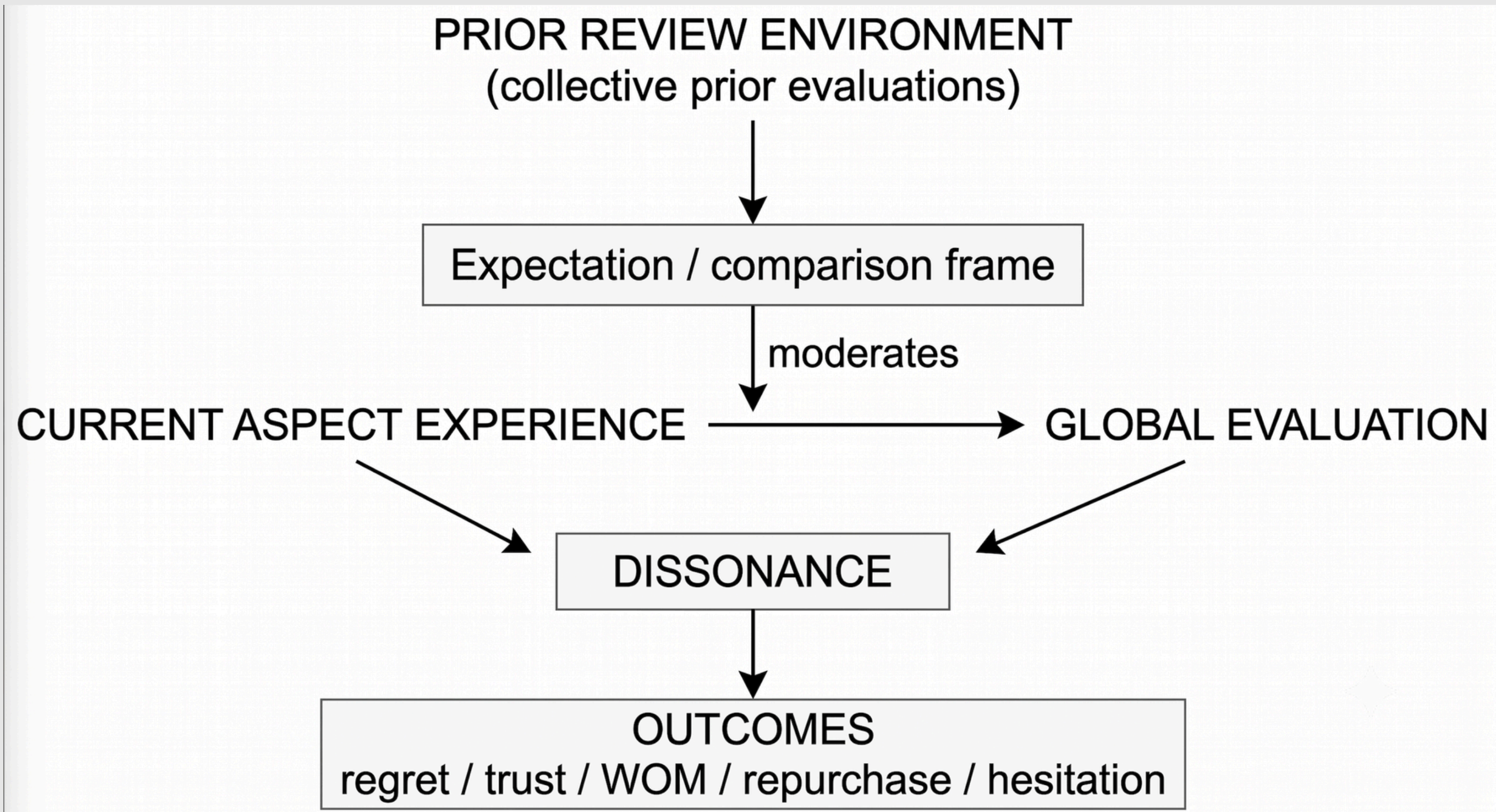
- TO EXAMINE HOW CONSUMERS' EVALUATIONS OF INDIVIDUAL PRODUCT ASPECTS INFLUENCE THEIR OVERALL/GLOBAL EVALUATION OF THE PRODUCT.
- TO INVESTIGATE HOW THE PRIOR ONLINE REVIEW ENVIRONMENT SHAPES CONSUMERS' INTERPRETATION OF THEIR CURRENT ASPECT-LEVEL EXPERIENCES.
- TO EXAMINE WHETHER AND HOW THE PRIOR REVIEW ENVIRONMENT MODERATES THE RELATIONSHIP BETWEEN CURRENT ASPECT-LEVEL EVALUATIONS AND CONSUMERS' GLOBAL PRODUCT EVALUATIONS, PARTICULARLY WHEN THERE IS A MISMATCH BETWEEN PRIOR REVIEWS AND CURRENT EXPERIENCE.



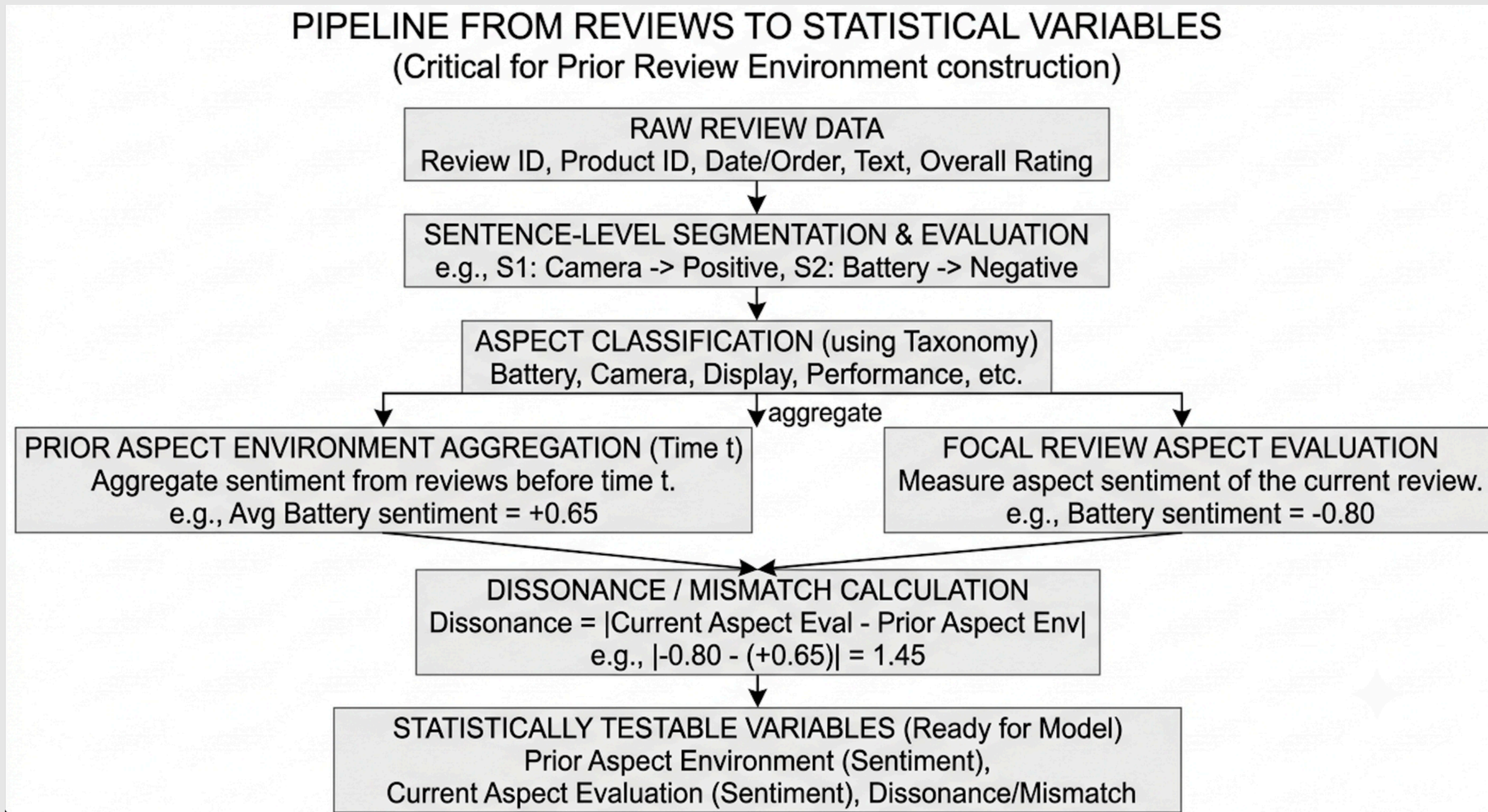
Proposed Architecture and Methodology (30% completed)

- Prior Review Environment Analysis – Analyze earlier reviews, ratings, and sentiment.
- Aspect Extraction & Sentiment – Identify product aspects and evaluate consumer sentiment toward each aspect.
- Global Evaluation Analysis – Extract the consumer's overall product evaluation from ratings and review text.
- AGED Detection & Temporal Analysis – Measure aspect–global dissonance and examine how it changes with the prior review environment over time.

Proposed Architecture and Methodology (30% completed)



Proposed Architecture and Methodology (30% completed)



Results(expected), Discussion & Comparison

Expected Results	Discussion	Comparison
AGED Detected	Contextual Effects	Beyond Static
Prior Influence	Expectation Effects	Beyond Influence
Temporal Changes	Dynamic Evaluation	Beyond Cross-Sectional
Better Prediction	Consumer Insights	Integrated Framework

Conclusion

- Identifies AGED as a distinct form of divergence between aspect-level and overall product evaluations.
- Highlights prior review influence on how consumers interpret their current product experiences.
- Captures temporal changes in consumer evaluations as the review environment evolves.
- Provides a dynamic framework for understanding online consumer decision-making.

Limitations & Future Work

Limitations	Future Work
Aspect Detection	Better NLP Models
Context Ambiguity	Context-Aware Analysis
Temporal Bias	Longer Timeframes
Platform Dependence	Cross-Platform Validation
Causality Limits	Experimental Validation

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